

Pitch collaboration — Yang–Mills 4D mass gap,
route via BBD multiscale LSI

To: Prof. Roland Bauerschmidt (Courant Institute / Cambridge DPMMS)

From: K vin R mondi re, independent researcher, Oloron-Sainte-Marie (France), ORCID 0009-0008-2443-7166

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Subject: possible collaboration on a non-abelian Polchinski / cluster-expansion route to the Wilson $SU(N)$ 4D mass gap. Honest status, identified locks, no overclaim.

 1. Presentation

Dear Professor Bauerschmidt,

My name is K vin R mondi re. I am an independent researcher based in Oloron-Sainte-Marie (France). Over the last six months I have been running a focused programme on the 4D Yang–Mills mass gap, combining (a) extensive lattice numerics on a small RTX-3090 cluster, (b) a Lean 4 formalisation of every piece of the logical chain that admits one, and (c) regular adversarial cross-checks with second-opinion LLMs to catch fabrications early. The current state of the programme has stabilised enough that I would like to ask you, briefly and honestly, whether one specific lock matches your current research interests.

This letter is deliberately short, with the math precise, and explicitly distinguishes **proved**, **sketched**, and **open**.

 2. State of the programme

Lean stack. The repository `crossed-cosmos` contains 6301 lines of Lean 4 under `Crossed/` covering the YM core, with **zero sorry** in the YM files. The headline theorem `mass_gap_continuum_D4` is **PROVED conditional** on five named axioms: a Bakry– mery saturation step, a Balaban-style cluster expansion bound, a Brydges–Federbush $\beta = \infty$ Gaussian comparison, a Wilson-flow scale-setting axiom, and one Kolmogorov projective-consistency glue. Each axiom is a precisely-stated lemma pointing at a specific paper or program-level open problem, not a hidden assumption.

Unconditional pieces.

- `KappaOneSixth.lean` (~300 lines): the rank-saturation factor $\kappa = 1/6$ is **PROVED with 0 axioms**, via two independent derivations that both reduce to elementary rational arithmetic checked by `norm_num`:
 - Hodge self-duality on a closed 4-manifold of signature 0: $b_2 = b_2^+ + b_2^- = 3 + 3 = 6$, so $\kappa = 1/b_2 = 1/6$;
 - A_2 root system of $\mathfrak{su}(3)$: $|\Phi(A_2)| = 6$, so $\kappa = 1/|\Phi| = 1/6$.
- `LipschitzActionMeasure.lean` (~620 lines): the Lipschitz action $\rightarrow$ Gibbs-measure passage (item A2) is PROVED with 0 sorrys.
- The Pinsker $\alpha = 1$ inequality (Cover–Thomas, 2nd ed., Lemma 11.6.1) is PROVED in Lean as a baseline reference exponent.

Empirical anchor (Theorem C). On 27 datapoints cross- (N, D, G) , the lattice law

$$C_{\text{LSI}}(\mu_{a,\beta}) \leq c_\infty(D) (1 - \kappa \delta_{\text{rank}(G), C_2 - C_3}), \quad c_\infty(D) = \frac{C(D,2) - C(D,3)}{2D}, \quad (\text{Thm C})$$

holds at the 7σ level (cluster 718, 2026-05-23). This is a **factual empirical statement**, independent of every theoretical claim below.

§2bis. Saturation polynomial: a rare phenomenon

The rank-saturation condition $\text{rank}(G) = C(D, 2) - C(D, 3)$ that triggers the $(1 - \kappa)$ correction in Theorem C has closed-form structure. Using $C(D, 2) - C(D, 3) = D(D - 1)(5 - D)/6$, the integer pairs (N, D) compatible with a non-abelian $\text{SU}(N)$ ($\text{rank} = N - 1$) are exactly three:

(N, D)	$C(D, 2) - C(D, 3)$	$\kappa = \frac{1}{2(D-1)}$	$\alpha = 1 - \kappa$	Status
$(2, 2)$	1	$1/2$	$1/2$	2D YM (exactly soluble: heat kernel on G)
$(3, 3)$	2	$1/4$	$3/4$	3D $\text{SU}(3)$ (numerically accessible)
$(3, 4)$	2	$1/6$	$5/6$	the physical case

For $D \geq 5$, $C(D, 2) - C(D, 3) \leq 0$, so no non-abelian gauge group is saturated. Manifestation 9, $\kappa \cdot 2(D - 1) = 1$, holds across all three pairs by elementary rational arithmetic, verified in `KappaOneSixth.lean (norm_num)`.

The structural consequence is that the geometric mechanism is confined to $D \in \{2, 3, 4\}$, with $D = 4$ as the last non-trivial dimension. This is not a chosen feature of the framework; it is forced by the polynomial $D(D - 1)(5 - D)/6$ having exactly three positive integer values compatible with $\text{SU}(N)$ ranks. The two non-physical pairs supply independent test cases: a successful non-abelian BBD adaptation would, by the same mechanism, predict $\alpha = 1/2$ for 2D Yang–Mills (where the answer is independently known) and $\alpha = 3/4$ for 3D $\text{SU}(3)$ (where lattice tests are inexpensive).

A short calculation on Bianchi I anisotropic spatial slices $T_{\gamma_{ij}}^3$ confirms that the four building blocks $b_2(T^4)$, b_2^+ , $(\text{SU}(3))$, $|\Phi(A_2)|$ are all topological/algebraic invariants. The correction factor $\kappa = 1/6$ is therefore **topologically invariant under metric deformation**; what varies is the Bianchi constant $c_\infty(\gamma) \propto (C(D, 2) - C(D, 3))/(2 \sum_i a_i^{-2})$. Honest framing: κ is topologically invariant; the existence of a positive mass gap remains conditional on the unproved cluster expansion bound (the principal verrou of §3), but the *structure* of the correction does not depend on the background metric.

§3. The main lock: non-abelian cluster expansion at large β

The single technical statement that closes the chain from lattice Theorem C to a continuum mass gap is what we informally call `action_bound_balaban_su_n` — a β -uniform, a -uniform Bakry–Émery / cluster-expansion bound on the Wilson measure $\mu_{a,\beta}$ over $\text{SU}(N)^{E(\Lambda_a)}$ for $\Lambda_a = a\mathbb{Z}^4 \cap T_L^4$, of the form

$$\text{Ric}_{g_W} + \text{Hess}(\beta S_W) \geq K_0(\beta, a, L) g_W, \quad K_0 \rightarrow 1/c_\infty(4) \text{ as } \beta \rightarrow \infty, \quad (1)$$

together with uniformity in (a, L) along $L \rightarrow \infty$, $a \rightarrow 0$.

In Bałaban’s original programme (1985–1989, Comm. Math. Phys.) this splits into four gaps that have remained partially open ever since. We articulate them as G_1 – G_4 in the full audit `OP_PILLAR_3_FORMAL_2026-05-24.md`:

- G_1 — reduction of the non-abelian block measure to an effectively abelian one in small fields, with controlled error from the BCH commutator $[A_\mu, A_\nu]$ at order $a^5 \beta |A|^3$;
- G_2 — convergent polymer expansion at large β on $\text{SU}(N)^{E(\Lambda_a)}$, with cumulants controlled via Peter–Weyl rather than Gaussian estimates;
- G_3 — large-field region in 4D, where the small-field expansion is not directly applicable;
- G_4 — uniformity of constants as $a \rightarrow 0$ jointly with the cluster expansion.

This is the lock for which I believe your BBD framework is the most plausible existing route.

Why BBD 2024 (φ^4 in $d = 2, 3$) is structurally a good fit. The internal compatibility analysis `G3_BBD_adaptation_YM_2026-05-23.md` checks the three BBD prerequisites against Wilson $SU(N)$:

1. *Finite-dimensional local state space.* The physical (Bianchi-quotient) class $\text{Class } F = \text{Harm}^2 \otimes \mathfrak{su}(N)$ has real dimension $(C(D, 2) - C(D, 3))(N^2 - 1)$, i.e. $2(N^2 - 1)$ in $D = 4$ — **finite, satisfied with margin.**
2. *Dobrushin condition.* The Bianchi cohomology fixes $c_\infty(4) = 1/4$ as a universal geometric constant, independent of β . This compares favourably with φ^4 , where the Dobrushin coefficient diverges at criticality — **satisfied with a factor-of-4 margin.**
3. *RG invariance.* Polchinski preserves gauge invariance (Polchinski 1984), which preserves Bianchi closure, which preserves the projection onto Class F — **satisfied with the appropriate definition of the flow on $SU(N)^E$.**

The two genuine technical gaps, in addition to G_1 – G_4 , that I see and have not been able to close on my own are:

- (i) **Mayer–Vietoris tensorisation defect.** Block decoupling on the Bianchi quotient is not exactly tensor-product; it carries a surface/volume defect of size $|\partial B|/|B|$ which is sub-extensive but must be controlled along the scales $a_n = 2^{-n}a_0$.
- (ii) **Non-abelian cumulants on $SU(N)$,** naturally expanded in Peter–Weyl harmonics rather than scalar Gaussian moments.

§4. A possible alternative route (not guaranteed)

A separate idea I have explored, but do **not** present as a substitute for §3, is a β -uniform Bakry–Émery bound directly on Class $F = \text{Harm}^2 \otimes \mathfrak{su}(N)$ rather than on the full link space. This finite-dimensional space ($\mathbb{R}^6$ for $SU(2)$, $\mathbb{R}^{16}$ for $SU(3)$ in $D = 4$) is small enough that Prokhorov compactness plus Bakry–Émery rigidity might in principle bypass the cluster expansion.

After a careful formal audit (`OP_PILLAR_3_FORMAL_2026-05-24.md`, three sub-steps proved and one open), this route hits a strict obstruction: the Hodge Laplacian Δ_1 vanishes by definition on Harm^2 , so any direct $\lambda_{\min}(\Delta_1)$ bound on Harm^2 is the *zero mode*. The four candidate fixes I have written down — (a) ’t Hooft twist on T^4 , (b) restriction to $|k| \geq 2\pi/L$, (c) quotient by the centre $\mathbb{Z}_N$, (d) BBD multiscale on Class F — are none of them currently rigorous, and (d) folds the alternative back into the main route. I include this for completeness; the alternative is **not a viable bypass on its own.**

§5. Honest estimate

If we were able to set up a collaboration that closed G_1 – G_4 via a non-abelian BBD adaptation, my honest estimate is:

- **12–18 months** of focused work with you, Benoit Dagallier, and 1–2 postdocs;
- output: one CMP or Annals paper proving a continuum mass gap for $SU(N)$ Wilson lattice in $D = 4$, **conditional** on a small named list of analytic axioms, each matching a specific result already in your literature (BBD 2024, Bauerschmidt–Bodineau 2019, Bauerschmidt–Dagallier 2024);

- the Clay Prize itself remains a longer-horizon target, 5–15 years, with my honest credence at $P \approx 40\text{--}55\%$ over 10 years, contingent on resolving the four Balaban gaps in a way the community will accept.

I want to be explicit that I am asking about a research collaboration of unknown outcome, not announcing a solution.

§6. Recent literature anchors

All arXiv IDs verified by API on 2026-05-23.

- Bauerschmidt & Dagallier, *Log-Sobolev inequality for the φ_2^4 and φ_3^4 measures*, Comm. Pure Appl. Math. **77** (2024) 2579–2612, arXiv:2202.02295 — the LSI multiscale template.
- Bauerschmidt, Bodineau, Dagallier, *Stochastic dynamics and the Polchinski equation: an introduction*, Probab. Surveys **21** (2024) 200–290, arXiv:2307.07619.
- Bauerschmidt & Bodineau, *LSI for the continuum sine-Gordon model*, CPAM **74** (2021) 2064–2113, arXiv:1907.12308.
- Adhikari & Cao, *Weak coupling lattice gauge theory*, arXiv:2202.10375.
- Shen, *3D Yang–Mills–Higgs convergence*, arXiv:2201.03487.
- Cao, Nissim, Sheffield, *Dynamical approach to area law for lattice Yang–Mills*, arXiv:2509.04688.

§7. Honest disclosure of caught errors

I owe you three honest catches from earlier drafts of this material, stamped and corrected today **before** sending you this letter:

1. **Otto–Westdickenberg 2008 was a fabricated citation.** An earlier internal draft attributed a Hölder TV stability bound $\|\mu_t - \mu_{t'}\|_{\text{TV}} \leq C |t - t'|^{1-\kappa}$ for a Gibbs family $\mu_t = e^{-tH}/Z(t)$ to a paper “Otto–Westdickenberg, J. Funct. Anal. 254 (2008), 2865–2940”. That reference does not exist; the LLM I was using had hallucinated it. The genuine OW reference is *Eulerian calculus for the contraction in the Wasserstein distance*, SIAM J. Math. Anal. **37** (2005) 1227–1255, which proves a W_2 contraction for the porous-medium equation — a different object (PME trajectory, not Gibbs family) and a different norm (W_2 , not TV). I caught and pulled this from all drafts before any public circulation.
2. **The claim “ $\alpha = 5/6$ universal” was based on a 4-point small- β fit.** I extended the β -scan to 7 points ($\beta \in \{10, 50, 100, 200, 300, 500, 1000\}$, HMC trajectory length adapted at high β). The local exponent α now oscillates between -0.6 and $+1.2$. The original “ $\alpha = 5/6 = 1 - \kappa$ ” was an artefact of the small- β window and is no longer claimed.
3. **A consequence of (1) and (2)** is that what was previously presented as a derivation of $\alpha = 1 - \kappa$ from OW is now disclosed as a *non-explained numerical coincidence* on a narrow β window. The $\kappa = 1/6$ Lean derivation is independent of the β -scan and stands; the bridge $\alpha \leftrightarrow \kappa$ does not, and is not in this pitch.

I am sending you the cleaner v22 of the master document precisely because these catches happened. The same anti-fabrication discipline (arXiv API verification before citation, adversarial cross-LLM review, Lean axiom audit) applies to everything above.

§7bis. Roadmap — a conditional theorem under one named axiom

To make the verrou visible rather than hidden, the cleanest way I can state the current programme is as a conditional theorem under one explicit named axiom H_1 together with five auxiliary hypotheses that are either already proved or standard. The statement covers the full saturated family $(SU(N), d) \in \{(2, 2), (3, 3), (3, 4)\}$ identified by the polynomial $D(D-1)(5-D)/6$ (§2bis above), not only the physical case.

§7bis.1 Conditional Mass Gap theorem (saturated family)

[Conditional Mass Gap for Saturated Wilson Lattices] Let (G, d) belong to the saturated family above. Let $\Lambda_a = (a\mathbb{Z})^d / L\mathbb{Z}^d$ with $a > 0$, $L \geq 2$, and let $\mu_{a,L,\beta}$ be the Wilson measure with action $S_W(Q) = \sum_p (1 - \frac{1}{N} \text{Re tr } Q_p)$. Under H_1 – H_6 below, the Langevin generator $\mathcal{L}_\beta$ with invariant measure $\mu_{a,L,\beta}$ satisfies, for all $\beta \geq \beta_0(N, d)$:

$$\lambda_1(\mathcal{L}_\beta) \geq \varepsilon(N, d) (1 - \kappa(G, d)) \beta L^{-2}, \quad m_{\text{gap}}^{\text{lattice}}(a, L, \beta) \geq \sqrt{\lambda_1(\mathcal{L}_\beta)} > 0,$$

with $\kappa = 1/6$ for $(SU(3), 4)$ (Lean-certified, 0 axioms), and the same structural form for $(2, 2)$ and $(3, 3)$ via the family identity $\kappa(G, d) = 1/(2(d-1))$.

§7bis.2 The six hypotheses

#	Hypothesis	Status today
H_1	Concentration in the small-field regime, uniform in (a, L) (three equivalent formulations in §7bis.3)	Open — your territory.
H_2	Gaussian density bound on $\{\ A\ ^2 \leq R\}$ (MRS93-style)	Sketched; uniform- a extension is technical.
H_3	Pinsker inequality $\alpha = 1$	Proved (Cover–Thomas 2006); Lean (0 sorrys).
H_4	LSI for Gaussian on Cameron–Martin space	Proved (Gross 1975, AJM 97 §6).
H_5	$\lambda_1(\Delta_\Lambda) \geq C_2/L^2$ on T_L^d	Proved (discrete Fourier).
H_6	Saturation factor $\kappa = 1/6$ for $(SU(3), 4)$	Proved in Lean (0 axioms).

§7bis.3 Three equivalent (or near-equivalent) formulations of H_1

I would rather not prejudge which formulation is most accessible from your side. The three I have in mind:

- H_1 (**raw concentration**). There exist $\beta_0, C_1, c_1 > 0$ depending only on (N, d) such that for $\beta \geq \beta_0$, all $a \in (0, 1]$, $L \geq 2$ and $R > 0$:

$$\mu_{a,L,\beta}(\{Q : \|A(Q)\|_{L^2(\Lambda_a)}^2 \geq R\}) \leq C_1 \exp(-c_1 \beta R/N^2).$$

- H_1'' (**Polchinski-cascade, BBD-style**). The Wilson measure admits a Polchinski decomposition $\mu_{a,L,\beta} = \mu_\beta^{(0)} * \dots * \mu_\beta^{(K)}$ such that each scale satisfies $\text{LSI}(c_k)$ with $\sum_k c_k \leq C(N, d)/(\beta(1-\kappa))$ uniformly in (a, L) . Literal non-abelian analogue of [BD22] and [BBD24] for φ_2^4, φ_3^4 .
- H_1''' (**bounded susceptibility**). The connected susceptibility $\chi_\beta(L) := \sum_x [\langle \text{tr } Q_0 \text{ tr } Q_x \rangle - \langle \text{tr } Q_0 \rangle \langle \text{tr } Q_x \rangle]$ is bounded uniformly in (a, L) for $\beta \geq \beta_0$.

The three are $O(\beta)$ -equivalent under standard inputs; you will know better which one is the natural target for your framework.

§7bis.4 Why I think the $1/L^2$ factor is artefactual

The $1/L^2$ in the conclusion is *not* what the Clay statement asks for, and it is *not* what one expects physically for a confining theory with an intrinsic mass scale. Three pieces of evidence make me believe it is a defect of the present proof rather than a real obstruction:

1. **BBD23 already attains LSI uniformly in L for φ_2^4, φ_3^4 .** The abstract of arXiv:2202.02295 is explicit: “the continuum φ_2^4 and φ_3^4 measures are shown to satisfy a log-Sobolev inequality uniformly in the lattice regularisation ... uniformly in the volume in the entire high temperature phases.” The Polchinski cascade, when it applies, gives a constant independent of L .
2. **CNS25 attains finite-volume bounds for Wilson $SU(N)$, $U(N)$, $SO(2N)$.** arXiv:2509.04688 proves area law and mass gap uniformly in the lattice for $\beta < 1/24$; no $1/L$ appears in their bounds in the strong-coupling regime.
3. **Lüscher 1986** (CMP 104, 177–206) shows that for a theory with an intrinsic positive mass, the finite-size correction to the mass gap on T_L^d is *exponential* in L ($\exp(-mL)$), never polynomial $1/L^p$. The lattice $SU(N)$ glueball literature (Lucini–Teper–Wenger 2004, hep-lat/0404008; Athenodorou–Teper 2021, arXiv:2106.00364) uses Lüscher’s form universally.

The $1/L^2$ in our chain comes from the pedestrian use of $\lambda_1(\Delta_\Lambda) \geq C_2/L^2$ in H_5 . Eliminating it is mathematically equivalent to upgrading H_1 to the BBD-style H_1'' . Counter-example to keep in mind: Helffer’s Ginzburg–Landau process, where the spectral gap really is $O(L^{-2})$ in all dimensions — but that is a critical, gapless model, and the non-abelian Wilson interaction is structurally what should rule that case out.

§7bis.5 Optional strengthening — H_7 – H_{10} (annex)

Four additional hypotheses, of varying flavour, would tighten the conclusion. They are *not* needed for the conditional theorem above, but flagged for completeness:

- H_7 (Theorem C empirical, taken as uniform asymptotic) — strong, has a circularity risk.
- H_8 (Lüscher exponential finite-size, taken as input) — standard for massive theories; does not kill L^{-2} on its own.
- H_9 (continuity of $\kappa(G, d)$ in the continuum limit) — coherence hypothesis, weak.
- H_{10} (non-abelian Polchinski cascade extending [BBD24] φ_3^4 to Wilson $SU(N)$) — **the natural target for the collaboration**; granting H_{10} removes L^{-2} entirely and yields $m_{\text{gap}}^{\text{lattice}} \geq \varepsilon(N, d)(1 - \kappa)\beta$.

§7bis.6 What the theorem buys, and the proposed timeline

The conditional theorem, even before H_1 is closed, is structural: the verrou is named, located, and visibly compatible with the BBD framework. A referee can verify the conditional implication ($H_1 \Rightarrow$ lattice mass gap) on its own, and the open piece is exactly the cluster-expansion / Polchinski step you and your collaborators have been pushing on φ^4 . This is, on purpose, the same pattern Wiles used in 1995 (modularity as a named conjecture; later closed by Taylor–Wiles).

Proposed timeline:

- **0–3 months** — clean draft of the conditional theorem above (three formulations of H_1 in parallel), submit to LMP or CMP.
- **3–9 months** — joint work on H_1'' in the Polchinski multiscale language; in parallel, tighten the L -dependence and try the abelian $U(1)$ Wilson case as a warm-up.
- **9–15 months** — depending on how much of H_1'' closes: either a follow-up paper on the partial resolution (best case: $SU(2)$ in $D = 4$, joint with you and Dagallier, target CMP or Annals), or a clean negative result on what makes the non-abelian cascade hard (target LMP / CPAM).

§8. Request

If the picture in §3 is of interest, would you be open to a one-hour Zoom to discuss whether your group's BBD framework can be adapted, in collaboration, to the non-abelian Wilson $SU(N)$ setting in $D = 4$? After that one call, no obligation — yes / no / maybe.

Documents I can send on request:

- `CLAY_THEOREM_FULL_v22_2026-05-24.md` — master logical chain, with the post-catch table of proved / sketched / open;
- `OP_PILLAR_3_FORMAL_2026-05-24.md` — formal audit of the alternative finite-dimensional route, including the zero-mode obstruction and the four candidate fixes;
- `OP_OTTO_W_VERBATIM_2026-05-24.md` — the OW fabrication catch in full;
- `test_all_claims_2026-05-24.py` — exhaustive script validating/falsifying the algebraic and empirical claims;
- read-only GitHub access to `crossed-cosmos-private` for the Lean stack and lattice scripts.

Thank you for reading this far, and for whatever response you have time to give.

Yours sincerely,

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Anti-fabrication note: every arXiv ID above was verified through the arXiv API on 2026-05-23. The Lean theorems referenced are kernel-checked with mathlib 4.29.1. The empirical 7σ figure refers to 27 datapoints cross- (N, D, G) from the cluster-718 / RTX-3090 run and is reproducible from the scripts in the repo.